Universal Scribe: Platform-Agnostic Identity & Anchoring Layer

Reference Specification — Version 1.0

Author: Leon Calvin Long II **Date:** July 2026

Abstract

Universal Scribe is a platform-agnostic identity and permanence layer designed to operate across heterogeneous blockchain systems. It provides deterministic capsule serialization, Merkle-based scribe trees, Taproot anchoring, EVM binding, Solana memo anchoring, and cross-chain orchestration flows. This specification defines the architecture, data formats, and anchoring mechanisms. Full reference implementations (TypeScript, Solidity, and pseudocode) are provided in an accompanying repository.

1. Introduction

Modern blockchain ecosystems lack a unified, portable identity layer capable of binding documents, metadata, and asset states across multiple chains. Universal Scribe addresses this gap by defining:

- deterministic identity capsules
- platform-neutral encoding
- Merkle scribe trees
- Bitcoin Taproot anchoring
- EVM binding interfaces
- Solana memo anchoring
- cross-chain orchestration flows

Universal Scribe is **vendor-agnostic**. It does not depend on any specific hosting provider or encoding scheme. Base44 is used in examples, but any encoding (Base58, Base64, Hex, Bech32) is supported.

2. Capsule Format (Platform-Agnostic)

A *capsule* is the atomic identity unit in Universal Scribe. Capsules serialize into a deterministic byte layout:

Field	Type	Description
version	uint8	Capsule version
hashAlg	uint8	Hash algorithm identifier
createdAtUnix	uint64	Creation timestamp
creatorPubkey	bytes32	Creator public key
contentHash	bytes32	Hash of referenced content
flags	uint16	Capsule flags
extraBytes	bytes	Optional metadata

Capsules produce:

- **uScribeId** — SHA-256 hash of the serialized capsule
- **identity string** — platform-neutral encoding (Base44 shown in examples)

Universal Scribe does not mandate any specific encoding.

3. Merkle Scribe Tree

A scribe tree is a Merkle structure composed of leaves:

- uScribeId
- optional binding payload

Leaf hashing uses a tagged SHA-256 domain (“ScribeLeafV1”). The Merkle root represents the entire scribe set and can be anchored on any chain.

This structure supports:

- batch identity proofs
- Taproot anchoring
- registry commitments
- cross-chain verification

4. Taproot Anchoring (Bitcoin)

Universal Scribe supports Bitcoin anchoring via Taproot tweaks:

1. Start with an internal x-only public key
2. Compute a tweak using BIP-340 tagged hashing
3. Add tweak·G to the internal key
4. Derive the Taproot output key

Anchoring the Merkle root inside the tweak binds the entire scribe tree to a single Taproot output.

This mechanism is compatible with:

- tiny-secp256k1
- bitcoinjs-lib
- rust-secp256k1

Pseudocode is provided in the reference repository.

5. Scribe Tree Manifest

A manifest describes the anchored scribe tree:

Field	Description
version	Manifest version (“scribe-tree-v1”)
taprootOutput	Bitcoin output reference (“txid:vout”)
internalKeyHex	Optional internal key
merkleRootHex	Merkle root
leaves	Array of leaf entries

Leaf entries include:

- uScribeId (hex)
- identity string (Base44 example)
- leaf hash
- position
- payload hash

Manifests can be stored on any platform:

- IPFS
- Arweave
- GitHub
- cloud storage
- enterprise registries
- Base44 (example)

Universal Scribe does not require any specific hosting provider.

6. EVM Binding & Registry

Universal Scribe provides Solidity interfaces for binding identities to smart-contract references.

IScribeBindable

Defines events and functions for binding/unbinding a uScribeId.

ScribeBindable

Base implementation with internal mapping and event emission.

ScribeRegistry

Global registry for recording bindings across contracts.

Scribe-Enabled ERC-20 Example

Demonstrates binding a uScribeId to a token reference.

These interfaces enable:

- asset provenance
- document binding
- audit trails
- cross-chain identity continuity

All EVM bindings are platform-neutral.

7. Solana Anchoring

Universal Scribe identities can be anchored using the Solana Memo program:

Code
```
US:<IdentityString>
```

The identity string may be Base44 (example) or any other encoding.

This provides a lightweight, immutable reference detectable by indexers.

8. Cross-Chain Orchestration Flow

A typical Universal Scribe workflow:

1. **Create capsule** → produce uScribeId + identity string
2. **Add to scribe tree** → compute leaf hashes
3. **Compute Merkle root**
4. **Anchor via Taproot tweak (optional)**
5. **Bind on EVM chain**
6. **Anchor on Solana via memo**
7. **Export manifest to any hosting platform**

This provides a unified identity layer across:

- Bitcoin
- EVM chains
- Solana
- off-chain systems
- any hosting platform

9. Example Encoding Module (Base44)

Base44 is used here **only as an example implementation**. Universal Scribe supports any encoding scheme.

Base44 provides:

- compact identity strings
- human-friendly characters
- deterministic encoding/decoding

Full Base44 reference code is included in the repository.

10. Reference Implementation Repository

Full code listings are provided externally to keep this specification concise.

The repository includes:

- TypeScript capsule library
- Merkle tree implementation
- Taproot tweak pseudocode
- Scribe manifest schema
- Universal Scribe orchestration module
- Solidity interfaces and contracts
- Solana memo anchoring example

This separation follows standard publication practice:

- **Specification in the paper**
- **Reference implementation in the repository**

11. Design Goals

Universal Scribe is designed to be:

- **platform-neutral**
- **vendor-agnostic**
- **interoperable**
- **portable**
- **future-proof**
- **compatible with any encoding or hosting provider**

Base44 is used only as an example.

12. Conclusion

Universal Scribe provides a unified identity and permanence layer capable of operating across multiple blockchain ecosystems. By separating the specification from the implementation and maintaining platform neutrality, Universal Scribe enables broad adoption, flexible deployment, and long-term interoperability.

Keywords

Identity, Capsule, Merkle Tree, Taproot, EVM, Solana, Interoperability, Blockchain, Cross-Chain, Anchoring, Registry, Platform-Agnostic.